ADITYA TALEKAR

AI Engineer — Agentic AI & MLOps

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EDUCATION

Master of Science in Data Science and Artificial Intelligence *Sep 2024 – Sep 2025*
Queen Mary University of London, London, UK — Distinction

Bachelor of Engineering in Civil Engineering *Apr 2021*
Savitribai Phule Pune University, Pune, India

EXPERIENCE

Assistant Professor — Artificial Intelligence *Jun 2026 – Present*
Dr. D. Y. Patil Pratishthan's College of Engineering, Kolhapur, India

- Teaching C Programming, Data Structures and Artificial Intelligence to about 300 students; the AI course runs from core theory (intelligent agents, search, reasoning under uncertainty, machine learning) to hands-on deployment of LLM-powered agents using vLLM, prompt engineering and GitHub-based workflows.
- Mentored student teams through AI agent hackathons, teaching them to provision AMD cloud droplets, generate API keys and self-host LLMs – demonstrated by deploying a 30B-parameter model myself and connecting it to a live application. Also serving as Industry Outreach Coordinator, among other additional responsibilities.

Independent AI Consultant (Freelance Work) *May 2026 – Jun 2026*
Exactitude Consultancy, Pune, India (Remote)

- Delivered a revamped website (in PHP) and an autonomous agentic system to automate workflows for a market research firm, collaborating with cross-functional stakeholders to translate business requirements into seven coordinated agents (LangChain, LangGraph).
- Configured custom Cloudflare rules, rate limiting and timeouts to block AI-scraper bots and automated scanners – defending against 1.35M threat-flagged requests (8% of 17.6M) in 48 hours.
- Automated blog/press-release drafting, LinkedIn and industry-segmented email outreach, with automatic failover across multiple LLM providers – scaling client outreach 10x, from 100 to 1,000, and cutting manual customer search from 10 hours to about 1 hour.
- Load-balanced across 10 free-tier LLM providers (including Gemini, Groq, NVIDIA NIM and Mistral) on GitHub Actions, rotating 33 API keys to cut external AI/data costs from an estimated $24,000–$36,000 to effectively $0.
- Engineered a data-quality pipeline in Python and MySQL using citation-grounded generation to curb hallucinated statistics, cutting chart-data mismatches from 39% to 0% (verified sample) and correcting 168 reports with wrong market-size figures.

Project Engineer *May 2021 – Jun 2024*
Talekar Infra LLP, Pune, India

- Automated cost, quantity and progress reporting for infrastructure projects using Python, Excel and Power BI dashboards – cutting report turnaround from 1 month to 1 week while managing a team of 10.
- Applied LLMs and Web Search to accelerate plan design and to source new clients, saving approximately 2 hours of manual work daily.

PROJECTS

NexusRAG (Enterprise RAG Platform) *Apr 2026 – May 2026*

- Built in agile sprints on FastAPI, PostgreSQL and pgvector, fusing BM25 lexical and dense-vector retrieval into top-50 candidates re-ranked to top-10 with BERT for citation-grounded responses over PDF documents, with session memory.

Multi-Agent System (Wave Engine) *Jan 2026 – Mar 2026*

- Shipped a self-correcting AI coding assistant on the Microsoft VS Code Marketplace (30+ active users) that checks and fixes its own output before returning it to the user.
- Coordinated 107 specialist agents through a local Qwen 3B orchestration model, quantised from 16-bit to 4-bit for speed at some accuracy cost – an agentic council verifies each result.
- Grounded code generation in a per-project knowledge graph and cut latency on most prompts from about 40s to 5s through prompt caching and KV-cache reuse.

Guardian (Autonomous Multi-Agent Monitoring Platform) *Oct 2025 – Dec 2025*

- Developed an AI system to solve a critical enterprise security problem — automatically discovering and monitoring an organisation's external infrastructure and flagging risks without human input.
- Built as a five-stage multi-agent reasoning pipeline on Neo4j and PostgreSQL – blackboard coordination, six reasoning strategies, episodic memory – over a knowledge graph spanning 960+ nodes and 1,764 relationships.
- Mapped 500+ endpoints in a single 10-minute scan of Bank of Japan's public-facing infrastructure – work that takes hours to days manually – uncovering hidden, internal and third-party-dependency endpoints and showing where the infrastructure is strong and weak.
- Also built an autonomous outreach agent that sourced 2,000+ investor and client leads and sent up to 200 personalised pitch emails a day, automating Guardian's business development end-to-end.

EnergyNexus — Energy Demand, Price & Wind Forecasting *Sep 2025*

Python, CNN-LSTM, Wavelet-LSTM, SQL, Time-Series Forecasting [GitHub]

- Built an end-to-end forecasting platform on two years of London energy and German renewable-energy data with CNN-LSTM and Wavelet-LSTM models, achieving R^2 of 95.9% (demand), 93.1% (wind) and 70.3% (price).

TECHNICAL SKILLS

Languages	Python, SQL, C
GenAI & LLMs	RAG, Multi-Agent Systems, Agentic Pipeline Architecture, LangChain, LangGraph, LLM Orchestration, vLLM, Prompt Caching, Vector Search, Hallucination Mitigation, Prompt Engineering
Machine Learning	PyTorch, TensorFlow, scikit-learn, Keras, Deep Learning, Distributed Training, NLP
Data & Infrastructure	PostgreSQL, MySQL, pgvector, FAISS, Neo4j, MongoDB, Power BI, FastAPI, Docker, Git, GitHub Actions, CI/CD, Cloudflare, AMD Developer Cloud